Probability Midterm 1: Makeup

50 minutes • 100 points • Six problems

Name: _____ **Date:** _____

Instructions. Answer all six problems and show your reasoning. Exact answers are acceptable; binomial coefficients need not be evaluated. No calculator is needed. Use the spaces provided; clearly label any work continued on the back. In Problem 6, use the guided steps rather than quoting a coupon-collector formula. Reserve time to check your work.

1. Sets — 10 points

Let $U = \{1, 2, \dots, 12\}$, $A = \{1, 3, 5, 7, 9, 11\}$, and $B = \{1, 2, 3, 4, 5\}$. Take all complements relative to U.

(a) Find $A \cap B$, $A \setminus B$, and $(A \cup B)^c$. (6 points)

(b) For these sets A, B, is the following statement true for every $C \subseteq U$? Justify with a proof or a counterexample. (4 points)

$$C \subseteq A \cup B \quad \Longrightarrow \quad C \subseteq A \text{ or } C \subseteq B.$$

2. A probability space — 10 points

Let $\Omega = \{a, b, c, d\}$, with every subset an event. The probabilities of a, b, c, d are $2k, 3k, 4k, 5k$, respectively.

(a) Find k, explaining the probability axiom you use. (4 points)

(b) Let $E = \{a, c\}$ and $F = \{b, c\}$. Find $\mathbb{P}(E)$, $\mathbb{P}(E \cap F)$, and $\mathbb{P}(E^c)$. (6 points)

3. Inclusion–exclusion **15 points**

In a group of 45 students, 20 play tennis, 17 play basketball, and 13 play volleyball. Eight play both tennis and basketball, six both tennis and volleyball, and five both basketball and volleyball. Three play all three. **The pairwise counts include students who play all three sports.**

(a) How many students play at least one of these sports? Use inclusion–exclusion. (11 points)

(b) How many students play none of these sports? (4 points)

4. Balls in an urn **20 points**

An urn contains six red, three blue, and four green balls. Four balls are selected uniformly without replacement: every four-ball subset of the thirteen individual balls is equally likely.

(a) Find the probability of selecting exactly two red, one blue, and one green ball. (10 points)

(b) Find the probability of selecting at least one green ball. (10 points)

5. A density, its mean, and its variance

25 points

A continuous random variable X has probability density

$$f(x) = \begin{cases} cx^2, & 0 \le x \le 1, \\ 0, & \text{otherwise}, \end{cases}$$

where c is a constant. Show the integrals you use.

(a) Find c. (5 points)

(b) Compute $\mathbb{E}[X]$. (8 points)

(c) Compute $\mathbb{E}[X^2]$, and hence $\mathrm{Var}(X)$. (12 points)

6. Collecting five sticker types **20 points**

Each packet contains one sticker. There are five sticker types, each equally likely, and different packets have independent contents. Duplicates are possible. Starting with no stickers, let T be the number of packets opened up to and including the first packet that completes the collection of all five types.

(a) After exactly k distinct types have been collected, let p_k be the probability that the next packet contains a new type. Find p_0, p_1, p_2, p_3, p_4, and explain your reasoning. (6 points)

(b) Let W_k be the number of additional packets needed to go from exactly k collected types to $k+1$ collected types, including the packet containing the new type. Find $\mathbb{E}[W_k]$ for $k = 0, 1, 2, 3, 4$. (8 points)
You may use: A geometric waiting time that counts trials through the first success, with success probability p, has mean $1/p$.

(c) Write T as a sum of the waiting times and find $\mathbb{E}[T]$. Name the expectation rule used. (6 points)

End of paper. Check that you have answered all six problems.